A1 Week 2 Plan

General Carbon/Contracting Theory & Numerical Methods — closing the gap to a Management Science submission

Status. Week 1 delivered the spine: the capacity-substitution channel K*R > K°R, the exact net-carbon inequality (C), the distribution-free cutoffs, the closed-form thresholds (the FCL adoption threshold and the renewable-cost bounds, with the small-FCL widening), a 15/15 machine-precision audit, a 3-referee honesty pass, and a green four-role integration. An adversarial Management Science review puts us at ≈70–75% of the rigor bar. The gap is one keystone — a general-distribution, finite-optimum carbon theorem — plus robustness, a clean numerical narrative, and theory positioning.

Scope (hard wall — A1 only). Every task below is general theory or numerical-method work on the FCL carbon model. Out of scope: real-data calibration/magnitudes (A2); reliability, welfare, social-vs-private optimum (B1); flexibility continuum, AI, storage, multi-grid, dynamics (B2). Roles stay in their own lane.

Role 1 — Optimal Contract & General Carbon Theory (lead; carries the heaviest load)

Mandate. A1 mandate (1): existence + full characterization of the optimal demand-response contract under general distributions, with a necessary-and-sufficient emission-reduction condition. Turn the headline from "conditional for general F" into a proved, finite-optimum theorem — the title depends on it.

- **Existence, uniqueness & contract characterization.** Prove coercivity of profit Π and a primitive condition on F (e.g. log-concave density) under which the interior capacity optimum is unique; prove (not merely assert) that the contract d*(x)=min{(K_F−x)_+, K_M} is the unique optimal dispatch in the admissible policy class via the strict R>c ordering — together the full characterization, discharging the "interior optimum is assumed" gap and Condition 2 (H negative definite) from primitives.
- **Primitive shift-ratio condition.** Replace the endogenous-window "g(x)=F_{K_R}/f non-increasing" with a support-wide condition on F that implies it, and prove the implication.
- **Necessary-and-sufficient emission condition, general F (keystone).** Under that primitive plus a tail/skew condition, prove a sharp iff for carbon reduction at a finite interior optimum for any such F — upgrading condition (C) from an endogenous inequality and the current existence conjecture to a clean necessary-and-sufficient theorem in primitives. Exhibit a non-power-law F satisfying all three primitive conditions (log-concavity, shift-ratio, tail/skew) to certify the admissible class is non-vacuous and genuinely more general than power-law/uniform.
- **Complete the viability threshold.** Extend the C_R† window from the power-law family to general F (the strict fall occurs at a finite, interior K*R, not the K*R=∞ corner), and prove Δ(C_R) is single-crossing so {$\Delta \geq 0$}=(C_R°, C_R†] is a genuine interval — i.e. carbon falls iff C_R < C_R†.
- **Main theorem + re-scope.** State the headline theorem, align the abstract/title to exactly what is proved, and write the economic intuition at journal quality.

Role 2 — Solver & Numerical Methods

Mandate. A1 mandate (4): build and document the numerical solver for d*(x) and k* the group relies on; deliver one coherent, reproducible backbone demonstrating the theory at a finite interior optimum.

- **General solver for d*(x) and k*.** Finalize and document the joint-capacity / dispatch solver for any residual-demand distribution — the validated backend every role builds on.
- **One carbon-sign narrative.** Replace the scattered "2/32" counts with a principled sweep over Role 1's primitive condition on a curated family, showing carbon falls at a finite interior optimum.
- **Hessian / existence certificate.** A reproducible check that the Hessian H is negative definite on the interior path for Role 1's distribution class (backstops existence & uniqueness).

- **Analytic-cofactor solver + one-command reproducibility.** Replace finite-difference cofactors with analytic ones (removes the noisy knife-edge points); ship a single reproduce harness covering the 15 exact-formula checks plus the carbon sweep, with documented tolerances.

Role 3 — Comparative Statics & Robustness

Mandate. A1 mandate (2): prove, under general conditions, the correlation comparative static and that mining raises optimal renewable capacity; close the cross-distribution robustness layer and deliver the sharpness example.

- **Correlation & price comparative statics (general).** Prove the net-carbon sign is governed by the demand–supply association (the shift ratio g = E[S|x]); that mining raises optimal renewable capacity K*R > K°R (A1 mandate 2); and sign the monotonicities in the model's key prices — ∂(carbon effect)/∂R (computing margin), ∂/∂β (blackout penalty), ∂/∂p (retail price) — which a theory referee always asks for and no Week-1 result covers.
- **Cross-F least-favorable & robust certificate.** Sign the capacity covariation dK*/dF along the shape class U so the least-favorable result holds across F, making $C_R \le C\dagger R \Rightarrow \Delta \ge 0$ a genuine distributionally-robust certificate (or prove the obstruction cleanly).
- **Reversed-channel counterexample.** Construct a realized F with g increasing ⇒ K*R < K°R — the sharpness example Week 1 could not produce (an algebraic sign-flip is not a realized example).
- **Repair the limit-interchange.** Replace the E[fossil*] → 0 a.s.-corner step with a dominated-convergence / finite-K_R envelope argument.

Role 4 — Mechanism Design, Regimes & Positioning

Mandate. A1 mandate (3): strengthen the mechanism-design treatment (regulator as principal, utility as agent); plus regime consistency, an explicit limitations boundary, and theory positioning — so the four pieces read as one paper with no exposed flank.

- **Mechanism design (principal–agent, with IR & IC).** Cast the regulator as principal and the utility/miner as agent; write the agent's participation (IR) and incentive-compatibility (IC) constraints and prove the price-based program implements d* optimally and, where feasible, uniquely (not merely as the principal's first-best dispatch); state the conditions — price-taking / exogenous R — under which IR/IC are slack and the induced miner-demand result holds — A1 mandate (3).
- **Unify & complete the regime map.** One Assumption 1 (R>c+C_M) and one carbon decomposition (Role 1's two-force split) everywhere; fold Role 1's finite-optimum statement and the C_R boundedness axis into a single consistent regime theorem with the Assumption-1 necessity pattern.
- **Limitations & scope boundary.** State explicitly what is assumed and what is deferred (real-data calibration → A2; reliability/welfare/social-vs-private → B1; flexibility continuum/storage/multi-grid/dynamics → B2), so no referee can reject on an unaddressed omission rather than an A1 error.
- **Positioning, bibliography & assembly.** Write the related-work section on the methods A1 uses (monotone comparative statics, interruptible/demand-response economics, distributionally-robust analysis), add the bibliography, move all "verified to 10^–k" out of proof bodies, and assemble one manuscript with a single notation table.

Why this is complete (acceptance gate). The four roles map one-to-one onto A1's own four mandate items — optimal contract + necessary-and-sufficient emission condition (Role 1), numerical solver (Role 2), correlation comparative static + mining-raises-K_R (Role 3), mechanism design + miner-demand conditions (Role 4) — and additionally close every rigor gap an adversarial Management Science review raised: existence/uniqueness from primitives, a primitive (not endogenous) condition, a finite-optimum general-F theorem, single-crossing of the carbon set in C_R, a non-vacuous admissible class, comparative statics in the key prices R/β/p, an IR/IC mechanism

(not just first-best dispatch), the cross-F robust certificate, the realized counterexample, the limit-interchange repair, a single consistent regime statement, an explicit limitations boundary, and literature positioning. With all of it landed there is no A1-scope angle — theory, rigor, robustness, mechanism, consistency, or positioning — left to reject on; anything beyond is, by construction, A2 or B-stream work.

Sequencing. All four start now and share only notation + the Week-1 closed forms. Role 1 delivers two primitives early — the primitive shift-ratio condition (unblocks Role 2's sweep and Role 3's class U) and the existence condition (unblocks Role 3's covariation and Role 2's Hessian check). Roles 2–4 hand Role 4 their final statements for assembly. Order of resolution: Role 1 (keystone) and Role 4 (mechanism + consistency) are gating; Roles 2–3 turn acceptance into a strong one.